

Performance Testing

Date	4 JULY 2026
Team ID	SWUID2026215775
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Credit card approval prediction
Test Environment	Local system (Flask Application)
Test Date	6-7-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a valid credit card application	10	60	Application should return prediction successfully
2	Multiple users submit applications simultaneously	20	60	System should handle concurrent requests without errors
3	Submit invalid or incomplete input data	10	30	System should display validation/error messages
4	Continuous prediction requests	25	120	System should maintain stable performance and quick responses

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	pass	Fast response time
2	Response Time (Max)	< 5 seconds	3.4 seconds	pass	Within acceptable limit
3	Throughput (Req/sec)	>10 Req/sec	15 Req/sec	pass	Good request handling
4	Error Rate	< 1%	0%	pass	No errors during testing
5	CPU Utilization	< 80%	55%	pass	Stable CPU usage
6	Memory Utilization	< 80%	62%	pass	Memory usage within limit

Step 4: Observations & Analysis

Key findings:

- The system provides accurate credit card approval predictions
- Average response time is under 2 seconds
- The application runs smoothly without major errors

Bottlenecks identified:

- Slight delay during initial model loading
- Prediction time increases with large input data
- Response time depends on server/internet speed

Optimization steps taken:

- Optimized data preprocessing for faster execution
- Used XGBoost model for efficient and accurate predictions
- Implemented input validation to reduce unnecessary errors

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

